report*

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1 Introduction

1) the key research question or knowledge gap that is addressed by the study;
2) the data used for this study;
3) the methods used to investigate the research question; and
4) the key findings and why they are important.

2 Data

The dataset comes from Inequality and Violent Crime: Evidence from Data on Robbery and Violent Theft by Eric Neumayer (Neumayer 2005) with the replication data found on Harvard Dataverse (Neumayer 2017). Each row represents a country during a specific year. The variables of interest include:

1. the Gini coefficient,
2. the ratio of the top to bottom quintile of the income distribution,
3. the log of the gross domestic product (GDP) per capita,
4. the growth rate of the GDP,
5. the unemployment rate,
6. the urbanization rate,
7. the female labor force participation rate,
8. the proportion of males between the ages of 15 and 64,
9. the Polity measure of democracy,
10. the human rights violation scale, and
11. the natural log of the number of robberies and violent thefts per million people which is the response in this study.

*Project repository available at: https://github.com/meganajoseph/pca_reg_comp.

The dataset is compiled from a variety of sources. The Gini coefficients come from the UN-WIDER database, and 6.6 is added to the values from expenditure rather than income (Neumayer 2005). The ratio fo the top to bottom quintile of the income distribution (data taken from A New Data Set Measuring Income Inequality by Deininger & Squire (Deininger and Squire 1996)) is used to measure income inequality(Neumayer 2005). The rest of the data are from the UN and the World Bank (Neumayer 2005).

4) highlight any critical characteristics of the data, possibly by using visualizations or summary tables.

3 Methods

1) clearly explaining the key distinctions between the univariate/multivariate methods;
2) explaining how you will compare the methods using your data; and
3) describing what you expect to see in your results (and why).

4 Results

The results section should describe the fitted models and the results of analysis. This section should include relevant tables, graphs, and quantitative results, as well as plain English explanations of all of these components, but should not include raw R output. Clearly explain what the results show and interpret them in the context of your research question

5 Discussion

briefly summarize your key findings, discuss potential weaknesses of your study, and identify potential improvements and extensions or future areas of related research

References

Deininger, Klaus, and Lyn Squire. 1996. "A New Data Set Measuring Income Inequality." *The World Bank Economic Review* 10 (3): 565–91. https://doi.org/10.1093/wber/10.3.565.

Neumayer, Eric. 2005. "Inequality and Violent Crime: Evidence from Data on Robbery and Violent Theft." *Journal of Peace Research* 42 (1): 101–12. http://www.jstor.org/stable/30042245.

Neumayer, Eric. 2017. *Replication Data for: Inequality and Violent Crime: Evidence from Data on Robbery and Violent Theft, Journal of Peace Research, 42 (1), 2005, Pp. 101-112.* Harvard Dataverse. https://doi.org/10.7910/DVN/6UAFWA.